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***B.Tech. Degree I Semester Regular/Supplementary &
II Semester Supplementary Examination March 2022***

**CE/EE/ME/SE 19-200-0101A & CS/EC/IT 19-200-0201B
COMPUTER PROGRAMMING
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) Mr. X wants to install some software into his new laptop. Which all types of software he needs? Explain with some examples.
- (b) Explain the steps in execution of a program with a neat diagram.
- (c) Predict the output of the following code:
- ```
void main ()
{
 int a =1,b=2,c=3;
 c=a++ + --b;
 printf("a=%d,b=%d,c=%d",a,b,c);
}
```
- Justify your answer.
- (d) What are identifiers? What are the rules for forming an identifier?
- (e) Write a program to demonstrate call by reference mechanism.
- (f) Explain any six string manipulation functions with their syntax.
- (g) "Arguments can be passed to the main function from command prompt". Comment on this statement with examples.
- (h) In a program two variables are declared and initialised as follows.
- ```
int a=5;
char c='A';
```
- Can a single pointer be used to point to both these variables? Justify your answer with examples.

PART B

(4 × 12 = 48)

- II. (a) Draw the block diagram of a digital computer. Explain each unit in detail. (8)
- (b) Give a brief description about different types of programming languages. (4)
- OR**
- III. (a) Write the algorithm and the flowchart for the program sum of digits of a number. (8)
- (b) List out and explain the steps involved in problem solving methodology. (4)
- IV. (a) Write a program to read a number and check whether the number is odd or even. If the number is even then print the square of the number and if the number is odd then print the cube of the number using the switch statement. (8)
- (b) Give any two types of iteration statements in C with syntax and example. (4)

OR

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- V. (a) Write a program to display the square of natural numbers upto a given number of terms and print their sum. (The series is as : 1 ,4 ,9, 16, ...,N²). (8)
 (b) Write the syntax for break and continue with examples. (4)
- VI. (a) Write a program to sort an integer array and print the second largest element in the array. Use a function for sorting. (8)
 (b) Explain any four differences between binary search and linear search. (4)
- OR**
- VII. (a) Create a structure Department with d_no, d_name and d_location. Write a program to read all the records and print them in lexicographical order of d_name. (8)
 (b) Differentiate between union and structure with examples. (4)
- VIII. (a) Write a program to print the sum of diagonal elements of a matrix after finding its transpose. Use only pointers to access the matrix elements. (8)
 (b) How can we use a pointer to allocate memory dynamically? Explain in detail with various functions used for it. (4)
- OR**
- IX. (a) A librarian wants to merge the contents of “Sports.txt” and “Arts.txt” into a file “Extracurricular.txt”. Write a C Program to implement the same. Also display the total number of characters, lines and spaces in the new merged file. (8)
 (b) Mr. X wants to do the following operations on the text file “new.txt” in the given order. (4)
 (i) display all details after the 10th character in the file
 (ii) display all characters again from the beginning
 (iii) display the current file pointer position.

Suggest the functions used for the above 3 operations with their syntax and use.
